

TESSERA R² 卡 R²

TECHNICAL REFERENCE MANUAL

技术参考手册

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LEGAL 法律

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INTRODUCTION 介绍

R² is the new receiver card for Tessera LED Processing. With a nominal 64K pixel capacity, R² is suitable for many types of panels, including very high resolution panels. R² also includes Tessera's full feature set - see www.bromptontech.com/tessera#features for details.

R² 是 Tessera LED 处理的新型接收器卡。R² 的标称像素容量为 64K，适用于多种类型的面板，包括非常高分辨率的面板。R² 还包括 Tessera 的完整功能集 - 有关详细信息，请参见 www.bromptontech.com/tessera#features。

At just 68x32mm R² is up to 6 times smaller than its predecessors and has just a single connection along the edge of the circuit board. It plugs into DDR2 SO-DIMM sockets that are widely and cheaply available. As such it can be integrated directly onto a panel's hub or spine board for the smallest and lowest-cost solution. Alternatively a simple carrier board can be used to adapt R² to match other common form factors for compatibility with existing designs.

R² 的尺寸仅为 68x32mm，比其前代产品小 6 倍，并且沿电路板边缘只有一个连接。它可插入广泛且价格低廉的 DDR2 SO-DIMM 插槽中。因此，它可以直接集成到面板的集线器或主干板上，以实现最小、成本最低的解决方案。或者，可以使用简单的载板来调整 R² 以匹配其他常见的外形尺寸，以便与现有设计兼容。

R² has been engineered for value and designed to remove unnecessary duplication of connectors and circuitry in a panel's design, giving a cost-effective overall solution.

R² 的设计具有价值，旨在消除面板设计中不必要的连接器和电路重复，从而提供具有成本效益的整体解决方案。

TERMINOLOGY 术语

- **Panel** - Single mechanical unit, usually consisting of several modules, a PSU, a controller card and external power and data connectors.
面板 - 单个机械单元，通常由多个模块、一个 PSU、一个控制器卡以及外部电源和数据连接器组成。
- **Module** or **LDM** (LED display module) - A single circuit board that houses LEDs and constitutes part of the face of the panel.
模块或 LDM (LED 显示模块) - 容纳 LED 并构成面板表面一部分的单个电路板。
- **Controller card** or **Receiver card** or **Card** - The card (R²) that is responsible for receiving Ethernet (or similar) data signals and distributing pixel data to the LED driver chips.
控制器卡或接收器卡或卡 - 负责接收以太网（或类似）数据信号并将像素数据分配到 LED 驱动芯片的卡（R²）。
- **Hub board** or **Spine board** - A simple board that is often used for signal distribution and to interface between the controller card and LED modules.
集线器板或主干板 - 一种简单的板，通常用于信号分配以及控制器卡和 LED 模块之间的接口。

MECHANICAL 机械

The Tessera R² card is compatible with the standard 200-pin DDR2 SO-DIMM form factor, designed to be installed in readily-available SO-DIMM sockets. All connections (including two Gigabit Ethernet interfaces, signals to the LED drivers, and user interface signals) are on the 200-pin gold-plated edge connector. A mounting hole is provided for additional mechanical security and thermal coupling. The top side of the R² card is shown in [Figure 1](#).

Tessera R² 卡与标准 200 针 DDR2 SO-DIMM 外形尺寸兼容，可安装在现成的 SO-DIMM 插槽中。所有连接（包括两个千兆以太网接口、LED 驱动器信号和用户界面信号）均在 200 针镀金边缘连接器上。提供了一个安装孔，用于额外的机械安全性和热耦合。R² 卡的顶面显示在

The card must be secured in its socket before power is applied. R² cards must not be hot-plugged.

在通电之前，必须将卡固定在其插槽中。R² 卡不得热插拔。

A clearance of at least 3.0mm is required on either side of the board, as shown in [Figure 2](#). Components should not normally be placed underneath the R² card on the carrier PCB. It is recommended that reverse-mounting connectors, which place the top side of the R² as shown in [Figure 1](#) face-down next to the carrier PCB, should not be used.

电路板两侧至少需要 3.0mm 的间隙，如图所示2卡下方。建议不要使用反向安装连接器，如图 1 所示 [class="fig_ref" title="图 1 - R² 顶视图俯视图" data-immersive-translate-walked="59903019-7bf0-413a-ad94-3bf8a89f1cc5">图 1](#) 正面朝下放在载波 PCB 旁边。

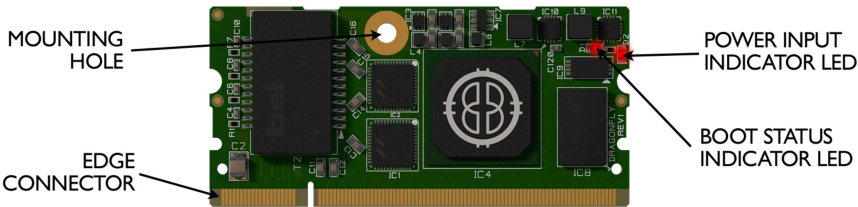


Figure 1 - R² top view
图 1 - R² 顶视图

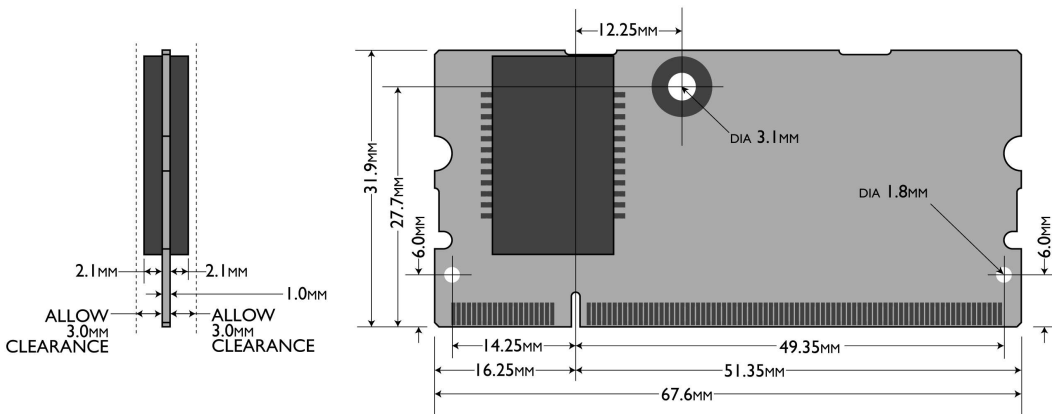


Figure 2 - Mechanical drawing
图 2 - 机械图

PIN OUTS 引脚输出

R² supports two different pin outs - LVCMOS mode and LVDS mode - which are described in [Table 3](#) and [Table 4](#) respectively. Note that pin roles from LVDS and LVCMOS modes cannot be mixed, pins should be used entirely in one mode or the other.

R² 支持两种不同的引脚输出 - LVCMOS 模式和 LVDS 模式 - 如表 4 - LVDS 模式引脚” [data-immersive-translate-walked="59903019-7bf0-413a-ad94-3bf8a89f1cc5">表 4](#)。请注意，LVDS 和 LVCMOS 模式的 pin 角色不能混合使用，pins 应完全在一种或另一种模式下使用。

LVC MOS Mode LVC MOS 模式	#	#	LVC MOS Mode LVC MOS 模式
do not connect 不连接	1	2	do not connect 不连接
do not connect 不连接	3	4	do not connect 不连接
do not connect 不连接	5	6	do not connect 不连接
ETH_0_SHIELD	7	8	ETH_1_SHIELD
ETH_0_SHIELD	9	10	ETH_1_SHIELD
do not connect 不连接	11	12	do not connect 不连接
do not connect 不连接	13	14	do not connect 不连接
do not connect 不连接	15	16	do not connect 不连接
ETH_0_MDI[0] +	17	18	ETH_1_MDI[0] +
ETH_0_MDI[0] -	19	20	ETH_1_MDI[0] -
do not connect 不连接	21	22	do not connect 不连接
ETH_0_MDI[1] +	23	24	ETH_1_MDI[1] +
ETH_0_MDI[1] -	25	26	ETH_1_MDI[1] -
do not connect 不连接	27	28	do not connect 不连接
ETH_0_MDI[2] +	29	30	ETH_1_MDI[2] +
ETH_0_MDI[2] -	31	32	ETH_1_MDI[2] -
do not connect 不连接	33	34	do not connect 不连接
ETH_0_MDI[3] +	35	36	ETH_1_MDI[3] +
ETH_0_MDI[3] -	37	38	ETH_1_MDI[3] -
do not connect 不连接	39	40	do not connect 不连接
	-		
	--	--	
do not connect 不连接	41	42	do not connect 不连接
do not connect 不连接	43	44	do not connect 不连接
do not connect 不连接	45	46	do not connect 不连接
do not connect 不连接	47	48	do not connect 不连接
GND	49	50	GND
ETH_1_LED_LINK	51	52	ETH_0_LED_LINK
ETH_1_LED_ACTIVITY	53	54	ETH_0_LED_ACTIVITY
GND	55	56	GND
do not connect 不连接	57	58	do not connect 不连接
FACTORY_BOOT_N	59	60	do not connect 不连接
GND	61	62	GND
I2C_1_SDA	63	64	I2C_0_SDA
I2C_1_SCL	65	66	I2C_0_SCL
GND	67	68	GND
V_OUT_2.5	69	70	V_OUT_2.5

V_OUT_2.5	71	72	V_OUT_2.5
GND	73	74	GND
I2C_INT	75	76	LED_DATA[0]
LED_DATA[32] <i>or</i> RESERVED_GPIO[5]	77	78	LED_DATA[1]
LED_DATA[32] 或 RESERVED_GPIO[5]			
GND	79	80	GND
LED_DATA[33] <i>or</i> RESERVED_GPIO[6]	81	82	LED_DATA[2]
LED_DATA[33] 或 RESERVED_GPIO[6]			
UI_TEST_BUTTON_N <i>or</i> RESERVED_GPIO[0]	83	84	LED_DATA[3]
UI_TEST_BUTTON_N 或 RESERVED_GPIO[0]			
GND	85	86	GND
RESERVED_GPIO[1]	87	88	LED_DATA[4]
UI_STATUS[1] <i>or</i> RESERVED_GPIO[2]	89	90	LED_DATA[5]
UI_STATUS[1] 或 RESERVED_GPIO[2]			
GND	91	92	GND
UI_STATUS[2] <i>or</i> RESERVED_GPIO[3]	93	94	LED_DATA[6]
UI_STATUS[2] 或 RESERVED_GPIO[3]			
UI_STATUS[3] <i>or</i> RESERVED_GPIO[4]	95	96	LED_DATA[7]
UI_STATUS[3] 或 RESERVED_GPIO[4]			
GND	97	98	GND
CAL_I2C_SPI[0]	99	100	LED_DATA[8]
CAL_I2C_SPI[1]	101	102	LED_DATA[9]
GND	103	104	GND
CAL_I2C_SPI[2]	105	106	LED_DATA[10]
CAL_I2C_SPI[3]	107	108	LED_DATA[11]
GND	109	110	GND
CAL_I2C_SPI[4]	111	112	LED_DATA[12]
CAL_I2C_SPI[5]	113	114	LED_DATA[13]
GND	115	116	GND
CAL_I2C_SPI[6]	117	118	LED_DATA[14]
CAL_I2C_SPI[7]	119	120	LED_DATA[15]
GND	121	122	GND
CAL_I2C_SPI[8]	123	124	LED_DATA[16]

CAL_I2C_SPI[9]	125	126	LED_DATA[17]
GND	127	128	GND
LED_SCAN[4]	129	130	LED_DATA[18]
LED_SCAN[0]	131	132	LED_DATA[19]
GND	133	134	GND
LED_SCAN[1]	135	136	LED_DATA[20]
LED_SCAN[2]	137	138	LED_DATA[21]
GND	139	140	GND
LED_SCAN[3]	141	142	LED_DATA[22]
LED_LATCH	143	144	LED_DATA[23]
GND	145	146	GND
LED_DATA[34] or RESERVED_GPIO[7]			
LED_DATA[34] 或 RESERVED_GPIO[7]	147	148	LED_DATA[24]
LED_DCLK_A	149	150	LED_DATA[25]
GND	151	152	GND
LED_DCLK_B	153	154	LED_DATA[26]
LED_DATA[35] or RESERVED_GPIO[8]			
LED_DATA[35] 或 RESERVED_GPIO[8]	155	156	LED_DATA[27]
GND	157	158	GND
LED_GCLK	159	160	LED_DATA[28]
BUFFER_OE_N	161	162	LED_DATA[29]
GND	163	164	GND
BUFFER_DIR	165	166	LED_DATA[30]
LED_DISCHARGE	167	168	LED_DATA[31]
GND	169	170	GND
VIRTUAL_FF_CLK_A	171	172	REBOOT_N
VIRTUAL_FF_CLK_B	173	174	do not connect 不连接
GND	175	176	GND
do not connect 不连接	177	178	DEBUG_V_OUT_1.8
do not connect 不连接	179	180	V_IN
GND	181	182	GND
do not connect 不连接	183	184	V_IN
DEBUG_CDI_TX	185	186	V_IN
GND	187	188	GND
do not connect 不连接	189	190	V_IN
do not connect 不连接	191	192	do not connect 不连接
GND	193	194	GND
do not connect 不连接	195	196	do not connect 不连接

do not connect 不连接	197	198	do not connect 不连接
GND	199	200	GND

Table 3 - LVCMOS mode pin-out
表 3 - LVCMOS 模式引脚输出

LVDS Mode is not yet available on general release. If you are interested in using LVDS mode in your project please contact us for further information. LVDS 模式在通用版本中尚不可用。如果您有兴趣在项目中使用 LVDS 模式，请联系我们以获取更多信息。 Table 4 - LVDS mode pin-out 表 4 - LVDS 模式引脚布局

SIGNAL DESCRIPTIONS 信号描述

[Table 5](#) below gives a brief description of the signals given in [Table 3](#) and [Table 4](#) above. These descriptions are not exhaustive - users should refer to the relevant sections and reference designs later in this document for further information.

表 3 - LVCMOS 模式引脚” LVCMOS 模式引脚 “data-immersive-translate-walked=”59903019-7bf0-413a-ad94-3bf8a89f1cc5“>表 3 和

All unused pins should be left unconnected unless otherwise specified.
 除非另有说明，否则所有未使用的 pins 都应保持未连接状态。

The following pins are used in both LVCMOS and LVDS mode pin-outs

以下引脚用于 LVCMOS 和 LVDS 模式引脚输出

Pin Name 引脚名称	I/O Standard I/O 标准	Description 描述
V_IN	Power input 电源输入	All pins labelled V_IN should be connected to the panel's power supply positive terminal (or a derivative thereof). See Table 6 for the recommended power supply range. 所有标有 V_IN 的引脚都应连接到面板的电源正极端子（或其衍生产品）。参见
GND	Common 常见	All pins labelled GND should be connected to the panel's power supply 0V terminal. 所有标有 GND 的引脚都应连接到面板的电源 0V 端子。
V_OUT_2.5	Power output 功率输出	These pins provide a reference voltage for LVCMOS outputs, for example to power the input-side of level shifting circuitry. Significant loads should not be pulled from these pins, see Electrical Specifications for details. If unused these pins should be left unconnected. 这些引脚为 LVCMOS 输出提供参考电压，例如为电平转换电路的输入侧供电。不应从这些引脚中拉出大量负载，有关详细信息，请参阅电气规格。如果未使用，则应保持这些引脚未连接。
do not connect 不连接	N/A 不适用	These pins should be left unconnected. They should not be tied to ground or a voltage rail. 这些引脚应保持未连接状态。它们不应连接到接地或电压轨。
ETH_x_SHIELD	Ethernet 以太网	Shield for Ethernet interface x. These should be connected to the shield on the connector used for ETH_x_MDI_[y]±. See Tessera Protocol Ethernet Ports section for details. 用于以太网接口 x 的盾牌。这些应连接到用于 ETH_x_MDI_[y]± 的连接器上的防护罩。有关详细信息，请参见 Tessera 协议以太网端口 部分。
ETH_x_MDI_[y]±	Ethernet 以太网	Ethernet Media Dependent Interface signals - pair y for interface x. These should be connected to external connectors to allow users to connect to a Tessera Processor and daisy-chain to other Tessera-enabled panels. All routing and connectors should be rated as suitable for Gigabit Ethernet. See Tessera Protocol Ethernet Ports section for details. 以太网媒体相关接口信号 - 接口 x 对 y。这些面板应连接到外部连接器，以允许用户连接到 Tessera 处理器，并以菊花链方式连接到其他支持 Tessera 的面板。所有路由和连接器都应被评为适用于千兆以太网。有关详细信息，请参见 Tessera 协议以太网端口 部分。

ETH_x_LED_LINK	2.5V output 2.5V 输出	Active-low 'link established' signal for Ethernet interface x - can optionally be connected to the cathode of an LED for status/debug purposes. Will be low when a Gigabit Ethernet link has successfully been established on the port. See Tessera Protocol Ethernet Ports section for details. 以太网接口 x 的“链路建立”信号 - 可以选择连接到 LED 的阴极，用于状态/调试目的。在端口上成功建立千兆位以太网链路时，将变为低电平。有关详细信息，请参见 Tessera 协议以太网端口 部分。
ETH_x_LED_ACTIVITY	2.5V output 2.5V 输出	Active-low activity signal for Ethernet interface x - can optionally be connected to the cathode of an LED for status/debug purposes. Will be low when Ethernet data is being transmitted or received on the port. See Tessera Protocol Ethernet Ports section for details. 以太网接口 x 的低电平有效活动信号 - 可以选择性地连接到 LED 的阴极，用于状态/调试目的。当端口上传输或接收以太网数据时，将变为低电平。有关详细信息，请参见 Tessera 协议以太网端口 部分。
FACTORY_BOOT_N	Open drain input 开漏输入	Hold this line low while the card is booting to force the card to boot into factory firmware (recovery) mode. It is strongly recommended that this signal is connected to a technician-accessible push button, but should not normally be user-accessible. It is strongly recommended that this is combined with the REBOOT_N signal - see Boot Process section for a reference design. If this pin is not used it should be left disconnected. 在卡启动时将此行保持为低电平，以强制卡启动到出厂固件（恢复）模式。强烈建议将此信号连接到技术人员可访问的按钮，但通常不应由用户访问。强烈建议将其与 REBOOT_N 信号结合使用 - 有关参考设计，请参阅 Boot Process 部分。如果未使用此引脚，则应将其断开连接。
REBOOT_N	Open drain input 开漏输入	Briefly pull this line low to make the card reboot. It is strongly recommended that this signal is connected to a technician-accessible push button, but should not normally be user-accessible. It is strongly recommended that this is combined with FACTORY_BOOT_N using an RC-timer circuit - see Boot Process section for a reference design. This pin is open-drain, so should not be driven high, only driven low or left floating. If this pin is not used it

		should be left disconnected.
		将此线短暂拉低以使卡重新启动。强烈建议将此信号连接到技术人员可访问的按钮，但通常不应由用户访问。强烈建议将其与使用 RC-timer 电路的 FACTORY_BOOT_N 结合使用 - 有关参考设计，请参阅 Boot Process 部分。该引脚是漏极开路，因此不应驱动为高电平，只能驱动为低电平或保持浮动状态。如果未使用此引脚，则应将其断开连接。
DEBUG_x	1.8V output 1.8V 输出	These pins are used for firmware debugging. To speed up panel development it is strongly recommended to connect this to a debug header - see Debug Header section for a reference design. These pins must only be used as described in the reference design, and otherwise left unconnected. 这些引脚用于固件调试。为了加快面板开发速度，强烈建议将其连接到调试头 - 有关参考设计，请参阅 Debug Header 部分。这些 pins 只能按照 reference design 中的说明使用，否则不得连接。
I2C_x_SDA, I2C_x_SCL I2C_x_SDA 、I2C_x_SCL	Open drain bidirectional. 开漏双向。	I2C buses for external devices. See Sensors, Fans and Monitoring section for details. These buses will typically be operated at 100kHz. These pins are open-drain, so should only be driven low or left floating. These pins should be pulled high with a 4K7 resistor, including if they are unused. 用于外部设备的 I2C 总线。有关详细信息，请参阅 传感器、风扇和监控 部分。这些总线通常以 100kHz 的频率运行。这些引脚是漏极开路的，因此只能被驱动为低电平或保持浮动状态。应使用 4K7 电阻器将这些引脚拉高，包括未使用的情况。
I2C_INT	2.5V input 2.5V 输入	Interrupt for external devices. See Sensors, Fans and Monitoring section for details. If this pins is unused it should be tied low. 外部设备的中断。有关详细信息，请参阅 传感器、风扇和监控 部分。如果此引脚未使用，则应将其绑低。
RESERVED_GPIO[x]	2.5V input/output 2.5V 输入/输出	These optional signals are used for panel-specific features. Please contact us for details. If unused these pins should be left unconnected. 这些可选信号用于面板特定的功能。请联系我们了解详情。如果未使用，则应保持这些引脚未连接。
UI_TEST_BUTTON_N	2.5V input 2.5V 输入	Active-low user interface test button signal. When held low for 3 seconds the controller card will enter/exit test pattern mode. Brief low pulses are used to change patterns while in test pattern mode. This pin can optionally be connected to a user-accessible push button. When not used this pin should be tied high. See User Interface Support

section for details.

低电平有效用户界面测试按钮信号。当保持低电平 3 秒时，控制器卡将进入/退出测试模式。在测试模式模式下，使用短暂的低脉冲来更改模式。该引脚可以选择连接到用户可访问的按钮。不使用时，此别针应系高。有关详细信息，请参阅 [User Interface Support](#) 部分。

User interface status indicator signals, for example for controlling status LEDs. See [User Interface Support](#) section for further details.

用户界面状态指示灯信号，例如用于控制状态 LED。有关更多详细信息，请参阅 [User Interface Support](#) 部分。

UI_STATUS[x]
2.5V
bidirectional.
2.5V 双向。

The following pins are used in LVCMOS mode pin-out only
以下引脚仅用于 LVCMOS 模式管脚

Pin Name 引脚名称	I/O Standard I/O 标准	Description 描述
CAL_I2C_SPI[x]	2.5V bidirectional. 2.5V 双向。	These signals can be used as an I2C or SPI interface to storage devices (such as EEPROM or flash) holding calibration data. See External Storage section for further details. 这些信号可用作存储设备（如 EEPROM 或闪存）的 I2C 或 SPI 接口，用于保存校准数据。有关更多详细信息，请参阅 External Storage 部分。
LED_DATA[x]	2.5V output 2.5V 输出	Data signals to be connected to the data input of the first LED driver IC in chain x. These signals may be used in any order, unused pins should be left unconnected. See Pixel Data Modes for details. 数据信号连接到链 x 中第一个 LED 驱动器 IC 的数据输入。这些 signals 可以按任何顺序使用，未使用的 pins 应保持未连接状态。有关详细信息，请参阅 Pixel Data Modes。
LED_SCAN[x]	2.5V output 2.5V 输出	Scan multiplex control signals, common to all driver IC chains. If fewer than the maximum number of LED_SCAN lines are required these signals should be used in ascending order. For example for 4:1 scan multiplexing, LED_SCAN[0] and LED_SCAN[1] should be used. Unused pins should be left unconnected. See Pixel Data Modes for details. 扫描所有驱动器 IC 链通用的多路复用控制信号。如果需要的 LED_SCAN 行数少于最大数量，则应按升序使用这些信号。例如，对于 4:1 扫描多路复用，应使用 LED_SCAN[0] 和 LED_SCAN[1]。未使用的引脚应保持未连接状态。有关详细信息，请参阅 Pixel Data Modes。
LED_LATCH	2.5V output 2.5V 输出	Data latch signal common to all LED driver IC chains. See Pixel Data Modes for details. 所有 LED 驱动器 IC 链共用的数据锁存信号。有关详细信息，请参阅 Pixel Data Modes。
LED_DCLK_x	2.5V output 2.5V 输出	Data clock signal. Where virtual channels are not being used LED_DCLK_A should be connected to all driver IC chains and LED_DCLK_B should be left unconnected. Where virtual channels are being used, both LED_DCLK_x signals are required - see Virtual Output Channels (LVCMOS Mode) for details. 数据时钟信号。在不使用虚拟通道的情况下，LED_DCLK_A 应连接到所有驱动器 IC 链，LED_DCLK_B 应保持未连接状态。当使用虚拟通道时，两个 LED_DCLK_x 信号都需要 - 有关详细信息，请参阅 虚拟输出通道 (LVCMOS 模式)。
LED_GCLK	2.5V output 2.5V 输出	Greyscale clock signal common to all driver ICs. When non-PWM-enabled driver ICs are used this signal acts as an active-low output

		enable signal. See Pixel Data Modes for details.
		灰度时钟信号是所有驱动器 IC 通用的。当使用非 PWM 使能驱动器 IC 时，该信号充当低电平有效输出使能信号。有关详细信息， 请参阅 Pixel Data Modes 。
LED_DISCHARGE	2.5V output 2.5V 输出	Sometimes known as <i>SR</i> or <i>Scan Reduce</i>. Active-high discharge signal, asserted during the period between scan bank switches. This signal can optionally be used to discharge power supplies on modules for ghosting elimination. 有时称为 <i>SR</i> 或 <i>Scan Reduce</i>。高电平有效放电信号，在扫描组开关之间的周期内置位。该信号可以选择用于对模块上的电源进行放电，以消除重影。
BUFFER_OE_N/DIR	2.5V output 2.5V 输出	Optional signals for controlling logic buffers connected to LED_* signals. See LVCMOS Mode for details. 用于控制连接到 LED_* 信号的 logic buffers 的可选信号。有关详细信息，请参阅 LVCMOS Mode。
VIRTUAL_FF_CLK_x	2.5V output 2.5V 输出	Flip-flop clock signals to be used in virtual channels mode. See Virtual Output Channels (LVCMOS Mode) section for details. If not using virtual channels, leave these pins disconnected. 在 virtual channels 模式下使用的 flip-flop clock signals。有关详细信息，请参阅 Virtual Output Channels (LVCMOS Mode) 部分。如果不使用 virtual channels，请断开这些 pins 的连接。

Table 5 - Signal descriptions

表 5 - 信号描述

ELECTRICAL SPECIFICATIONS

电气规格

Recommended Operating Conditions

建议的作条件

Symbol 象 征	Parameter 参 数	Min 最 小 值	Typ 类 型	Max 麦 克 斯	Units 单 位	Notes 笔 记
V_{IN}	Nominal power supply voltage on V_IN pins V_IN 引脚上的标称电源电压	3.3	5.0	5.0	V	Temporary $\pm 10\%$ variation from nominal min/max permitted. 临时 $\pm$ 允许的标称最小值/最大值的 10% 变化。
I_{IN}	Current consumption on V_IN pins V_IN 引脚上的电流消耗		1.0	TBC 待定	A	Measured at $V_{IN}=5.0$ V. 在 $V_{IN}=5.0$ V 时测量。
			TBC 待定	TBC 待定	A	Measured at $V_{IN}=3.3$ V. 在 $V_{IN}=3.3$ V 时测量。
$V_{OUT_2.5}$	Output reference voltage on V_OUT_2.5 pins V_OUT_2.5 引脚上的输出参考电压		2.5		V	
$I_{OUT_2.5}$	Permitted current consumption on V_OUT_2.5 pins V_OUT_2.5 引脚上的允许电流消耗			50	mA 马	
V_{IH_LVCMOS}	Input high voltage threshold on all LVCMOS inputs 所有 LVCMOS 输入上的输入高电压阈值	1.7			V	
V_{IL_LVCMOS}	Input low voltage threshold on all LVCMOS			0.7	V	

	inputs 所有 LVCMOS 输入上的输入 低电压阈值			
VOH _{LVCMOS}	High level output voltage on all LVCMOS output 所有 LVCMOS 输出均具有高 电平输出电压	2		V
VOL _{LVCMOS}	Low level output voltage on all LVCMOS output 所有 LVCMOS 输出上的低电 平输出电压		0.4	V
IOL _{LVCMOS}	Sink current on LVCMOS outputs (except ETH_x_LED_y pins) LVCMOS 输出 上的灌电流 (ETH_x_LED _y 引脚除外)		1.0	mA 马
IOH _{LVCMOS}	Source current on LVCMOS outputs (except ETH_x_LED_y pins) LVCMOS 输出 上的源电流 (ETH_x_LED _y 引脚除外)		1.0	mA 马
IOL _{ETH_LED}	Sink current on ETH_x_LED_y pins ETH_x_LED_y 引脚上的灌电 流		2.0	mA 马
t _{REBOOT_N}	REBOOT_N low pulse width REBOOT_N 低 脉冲宽度	2.0		μs 微 秒

Table 6 - Recommended electrical conditions

表 6 - 建议的电气条件

Absolute Max Conditions 绝对最大条件

Symbol 象征	Parameter 参数	Min 最小 值	Typ 类型	Max 麦克斯	Units 单位	Notes 笔记
V _{ILVCMOS}	Input voltage on all LVCMOS inputs	-0.3		3.6	V	

Table 7 - Absolute maximum electrical conditions

THERMAL CONSIDERATIONS

Like any electronic components, it is important that R² is not allowed to overheat. How this is achieved will depend largely on the panel's design and other components in the panel such as the PSU and driver ICs. The power consumption of the R² can be calculated from the values given in [Table 6](#). Panel designs therefore should take into account the need to dissipate this energy as heat.

To aid with thermal dissipation from the R² card itself, the card should be mounted to 0V reference plane on the carrier board using a metal spacer/fixings.

BOOT PROCESS 引导过程

When power is applied to V_{IN} and REBOOT_N is deasserted (pulled high) the card will begin the boot process.
当 V_{IN} 接通电源并且 REBOOT_N 置低（拉高）时，卡将开始引导过程。

First the Factory Firmware will be loaded. The Factory Firmware is programmed when the card is manufactured and cannot normally be updated by end-users. It contains a minimal set of features such as firmware reload, but cannot receive video data or illuminate panels. Normally the Factory Firmware will identify which of two User Firmware banks is the most recent and then proceed to reboot into that bank.

首先，将加载出厂固件。出厂固件是在制造卡时编程的，最终用户通常无法更新。它包含一组最少的功能，例如固件重新加载，但无法接收视频数据或照亮面板。通常，Factory Firmware 会识别两个用户固件库中的哪一个是最新的，然后继续重新启动该 bank。

In the unlikely event that a User Firmware bank becomes corrupted, users can force the card to remain in Factory Firmware and do the firmware reload from there. To do this, the FACTORY_BOOT_N should be held low for several seconds after REBOOT_N is deasserted. The boot status indicator led (see [Figure 1](#)) will flash to confirm that the card is forcibly remaining in Factory Firmware mode.

万一用户固件库损坏，用户可以强制卡保留在出厂固件中，并从那里重新加载固件。为此，FACTORY_BOOT_N 应在 REBOOT_N 置低后保持低电平几秒钟。引导状态指示灯指示灯（参见

It is strongly recommended that the REBOOT_N and FACTORY_BOOT_N pins are connected to a technician-accessible push button(s) to allow this factory recovery procedure to be carried out. These should not normally be accessible to end-users. The two signals can be combined using an RC-timer circuit where a brief press will reboot the card and a press-and-hold will reboot the card into factory firmware. For an example, see [Figure 8](#).

强烈建议将 REBOOT_N 和 FACTORY_BOOT_N 引脚连接到技术人员可访问的按钮，以便执行此出厂恢复程序。最终用户通常不应访问这些 Cookie。这两个信号可以使用 RC 定时器电路进行组合，短按将重新启动卡，按住将卡重新启动到出厂固件。有关示例，请参见

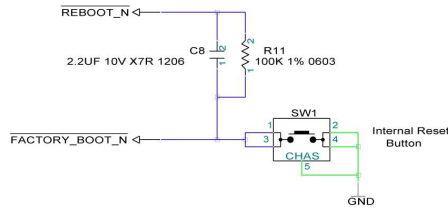


Figure 8 - Recommended reboot circuitry

图8 - 建议的重启电路

CONFIGURATION AND FIRMWARE RELOAD

配置和固件重新加载

When an R² card is first installed into a panel, it must first be programmed with a Configuration using the Tessera Configuration Application. The Configuration tells the card how to control the panel, and includes information such as the number of LEDs and type of LED driver. Configurations are currently written by us - see the [Design Process](#) section for details.

当 R² 卡首次安装到面板中时，必须首先使用 Tessera Configuration 应用程序通过配置对其进行编程。Configuration 告诉卡如何控制面板，并包括 LED 数量和 LED 驱动器类型等信息。配置目前由我们编写 - 有关详细信息，请参阅 [Design Process](#) 部分。

The Configuration is stored in non-volatile memory inside the R² card, so only needs to be programmed once when the card is first installed into a panel.

配置存储在 R² 卡内的非易失性存储器中，因此只需在首次将卡安装到面板中时进行一次编程。

Tessera is a continually-evolving system, and periodically we will issue updated firmware for receiver cards to add new features or address issues. End-users can easily update this firmware themselves using a Tessera Processor. The card will check the firmware update against its Configuration, so it is not possible for end-users to damage a panel by accidentally loading incorrect firmware or settings.

Tessera 是一个不断发展的系统，我们将定期发布接收器卡的更新固件，以添加新功能或解决问题。最终用户可以使用 Tessera 处理器轻松更新此固件。该卡将根据其配置检查固件更新，因此最终用户不会因意外加载不正确的固件或设置而损坏面板。

All firmware reloads are fail-safe. If the card loses power during a firmware upload, the new firmware will be discarded and the card will continue to use the existing firmware.

所有固件重新加载都是故障安全型的。如果存储卡在固件上传过程中断电，新固件将被丢弃，存储卡将继续使用现有固件。

DEBUG HEADER 调试标头

The R2 has a dedicated debug output pin, which is used for testing and debugging firmware or panel integration issues.

R2 具有专用的 debug 输出引脚，用于测试和调试固件或面板集成问题。

The DEBUG_CDI_TX and DEBUG_V_OUT_1.8V pins must be connected to an internal pin header as described in [Figure 9](#).

DEBUG_CDI_TX 和 DEBUG_V_OUT_1.8V 引脚必须连接到内部引脚接头，如

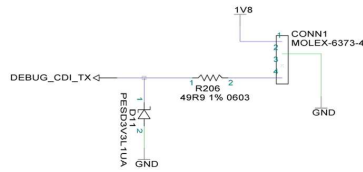


Figure 9 - Recommended debug header

图 9 - 建议的调试标头

TESSERA PROTOCOL ETHERNET PORTS

TESSERA 协议以太网端口

R² cards are connected to Tessera Processors by the Tessera Protocol, which operates on top of standard 1000BASE-T Gigabit Ethernet, as defined in IEEE 802.3. For optimum performance, any internal routing, cabling and connectors between the card's Ethernet pins (ETH_x_MDI[y]±) and the panel's external connectors should be standards-compliant.

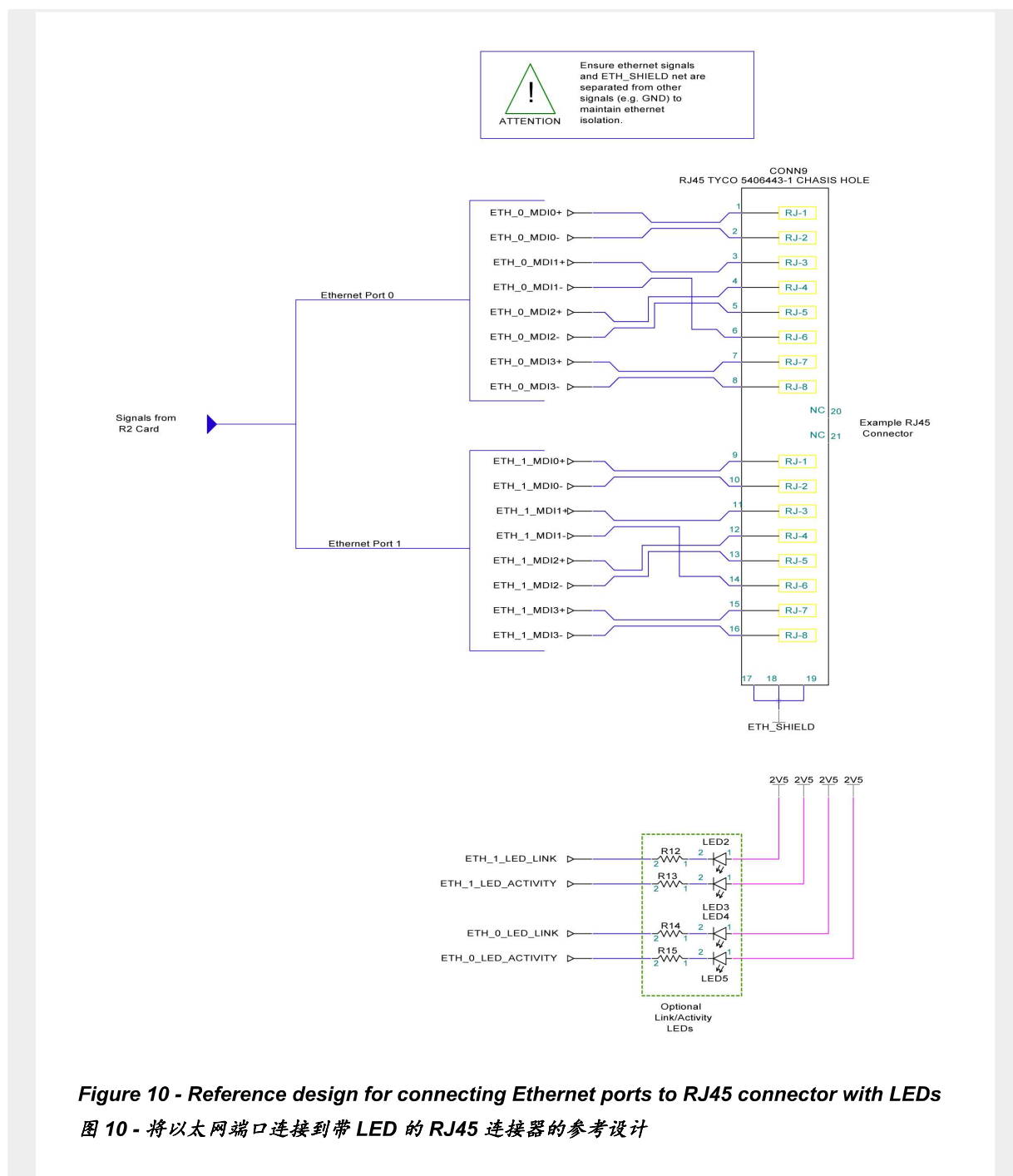
R² 卡通过 Tessera 协议连接到 Tessera 处理器，该协议在 IEEE 802.3 定义的标准 1000BASE-T 千兆以太网上运行。为了获得最佳性能，卡的以太网引脚（ETH_x_MDI[y]±）和面板的外部连接器之间的任何内部布线、电缆和连接器都应符合标准。

The two Ethernet ports on the R² - ETH_0_MDI[y]± and ETH_1_MDI[y]± - are identical and can be daisy-chained to other cards in either direction. There are no restrictions on the cabling topology that users use to connect panels to the processor. Tessera Processors provide a means of automatically detecting cabling topology - see Tessera system documentation for further details.

R² 上的两个以太网端口 - ETH_0_MDI[y]± 和 ETH_1_MDI[y]± - 是相同的，可以菊花链方式在任一方向连接到其他卡。用户用于将面板连接到处理器的布线拓扑没有限制。Tessera 处理器提供了一种自动检测布线拓扑的方法 - 有关详细信息，请参阅 Tessera 系统文档。

The ETH_x_MDI[y]± signals are isolated from the rest of the R²'s electronics, and this isolation should be maintained in the panel's other circuit boards. The ETH_x_SHIELD signals should be connected to the shield on the external Ethernet connectors.

ETH_x_MDI[y]± 信号与 R² 的其余电子设备隔离，这种隔离应在面板的其他电路板中保持。ETH_x_SHIELD 信号应连接到外部以太网连接器上的扩展板。



PIXEL DATA MODES 像素数据模式

A nominal 65536 pixels can be controlled by the LED outputs on the R². Most standard PWM-enabled or static constant-current chips can be supported. Support for new ICs can usually be added if a comprehensive datasheet is provided. See [Supported Driver ICs](#) for details.

标称 65536 像素可由 R² 上的 LED 输出控制。可以支持大多数标准 PWM 或静态恒流芯片。如果提供了全面的 Data 书，通常可以添加对新 IC 的支持。有关详细信息，请参阅[支持的驱动程序 ICs](#)。

Two signalling modes are supported - LVCMOS and LVDS.

支持两种信号模式 - LVCMOS 和 LVDS。

LVC MOS Mode LVC MOS 模式

In LVC MOS mode up to 36 chains of LED driver ICs can be connected directly to the LED_DATA[x] pins on the R². This can be extended to up to 72 chains using Virtual Channels (see section [Virtual Output Channels \(LVC MOS Mode\)](#) for details). These 72 chains are equivalent to '24 RGB groups' in other receiver cards. Note that pins LED_DATA[35:32] are shared with other functions, so if these functions are required, the number of chains should be limited to 32 or fewer (or 64 or fewer when using Virtual Channels).

在 LVC MOS 模式下，多达 36 链的 LED 驱动器 IC 可以直接连接到 R² 上的 LED_DATA[x] 引脚。这可以使用虚拟通道扩展到多达 72 个链（有关详细信息，请参阅 [虚拟输出通道（LVC MOS 模式）虚拟](#) 部分）。这 72 个链相当于其他接收器卡中的“24 个 RGB 组”。请注意，引脚 LED_DATA[35: 32] 与其他函数共享，因此如果需要这些功能，链的数量应限制为 32 个或更少（或使用 Virtual Channel 时应限制为 64 个或更少）。

Each chain of driver ICs shares a data clock (LED_DCLK_A) and latch (LED_LATCH). In the case of PWM-enabled ICs, the LED_GCLK pin provides a greyscale clock, for constant current ICs it provides an active-low output enable signal.

每个驱动器 IC 链共享一个数据时钟（LED_DCLK_A）和锁存器（LED_LATCH）。对于使能 PWM 的 IC，LED_GCLK 引脚提供灰度时钟，对于恒流 IC，它提供低电平有效输出使能信号。

There is no restriction on the arrangement of pixels or driver chips on each output channel. Designers are therefore free to interleave LEDs of different colours on the same driver IC chain, providing all ICs on a chain use the same configuration register settings. For best performance all used channels should drive the same number of ICs and pixels.

每个输出通道上的像素或驱动芯片的排列没有限制。因此，设计人员可以自由地在同一驱动器 IC 链上交错使用不同颜色的 LED，前提是链上的所有 IC 都使用相同的配置寄存器设置。为了获得最佳性能，所有使用的通道应驱动相同数量的 IC 和像素。

Scan multiplexing up to a 32:1 ratio is provided using the LED_SCAN[x] pins listed above. These signals should be considered as a 5-bit binary bus with SCAN[0] being the least significant bit. A binary decoder can be used in the panel to convert this binary bus into one-hot signals for enabling scan bank power supplies. It is recommended that the scan multiplexing ratio is as low as possible to avoid significantly reducing the refresh rate. As such, scan multiplex ratios above 16:1 are strongly discouraged.

使用上面列出的 LED_SCAN[x] 引脚提供高达 32: 1 比率的扫描多路复用。这些信号应被视为 5 位二进制总线，其中 SCAN[0] 是最低有效位。可以在面板中使用二进制解码器将此二进制总线转换为独热信号，以启用扫描组电源。建议扫描多路复用率尽可能低，以避免显著降低刷新率。因此，强烈建议不要使用高于 16: 1 的扫描多路复用率。

The LED_DISCHARGE is an active-high signal which is asserted during the scan bank switching period. This signal can optionally be used to control dedicated power supply discharge circuitry on LED modules for ghosting elimination.

LED_DISCHARGE 是一个高电平有效信号，在 scan bank 切换期间被置位。该信号可选地用于控制 LED 模块上的专用电源放电电路，以消除重影。

For driver ICs with internal configuration registers (for example those with programmable current gains), it is possible to specify a different value for the configuration register for each output channel. Different register values can not be programmed into different ICs on the same output channel.

对于具有内部配置寄存器的驱动器 IC（例如具有可编程电流增益的 IC），可以为每个输出通道的配置寄存器指定不同的值。不同的寄存器值不能编程到同一输出通道上的不同 IC 中。

Where it is necessary to fan-out signals to multiple modules or chains, for improved signal integrity it is recommended that the fan-out is placed immediately prior to a buffer or level-shifter. The outputs from R² should not be used directly to run over long traces/cables or multiple connectors. Instead appropriate buffers or level-shifters should be used. For best performance it is recommended that the same number of buffer or level-shifter stages are used on all signals in order to match the delay. For an example of level-shifting the R² outputs to 5V and fanning-out to multiple modules, see [Figure 11](#).

当需要将信号扇出到多个模块或链时，为了提高信号完整性，建议将扇出放在缓冲器或电平转换器之前。R² 的输出不应用于直接用于长走线/电缆或多个连接器。相反，应使用适当的缓冲区或电平转换器。为了获得最佳性能，建议在所有信号上使用相同数量的缓冲器或电平转换器级，以匹配延迟。有关将 R² 输出电平转换至 5V 并扇出至多个模块的示例，请参见

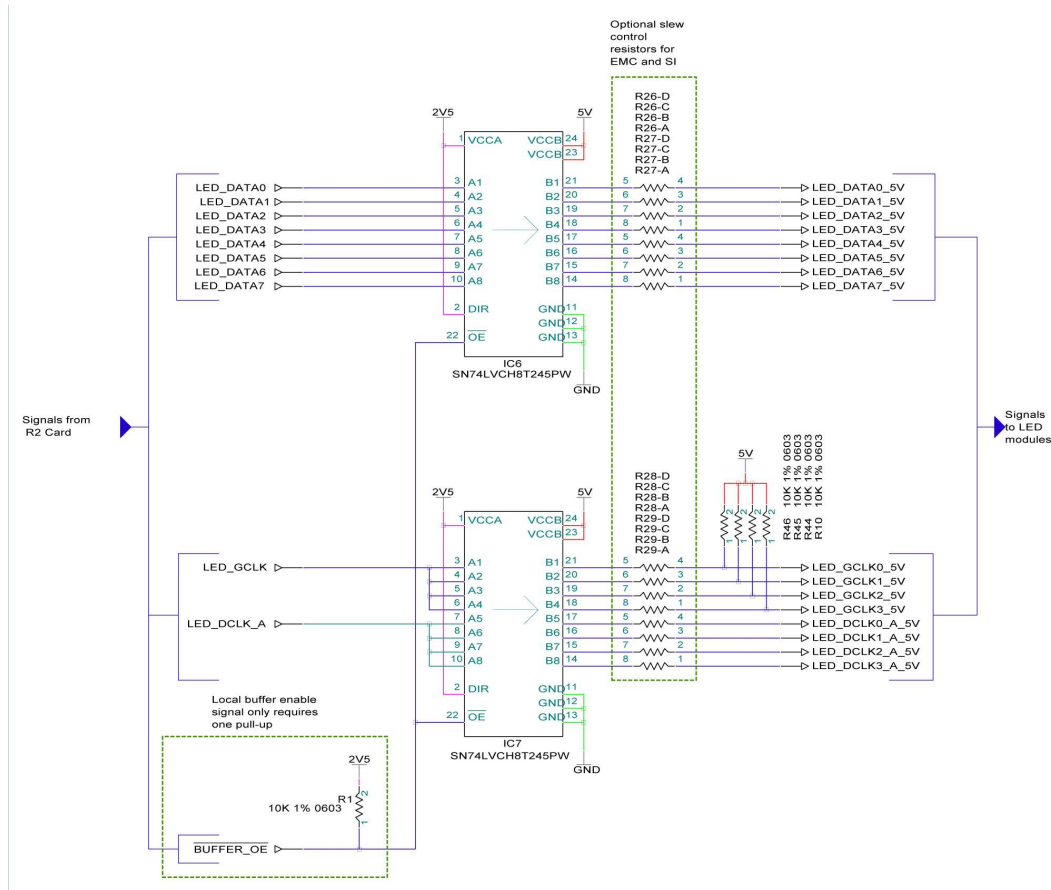


Figure 11 - Reference design for buffering and level shifting LED Driver signals to 5V

图 11 - 用于将 LED 驱动器信号缓冲和电平转换至 5V 的参考设计

LVDS Mode LVDS 模式

LVDS Mode is not yet available on general release. If you are interested in using LVDS mode in your project please contact us for further information.

LVDS 模式在通用版本中尚不可用。如果您有兴趣在项目中使用 LVDS 模式，请联系我们以获取更多信息。

Supported Driver ICs 支持的驱动程序 IC

The R² currently supports many types of PWM-enabled driver ICs in LVCMOS and LVDS modes, and most static-type ICs in LVCMOS mode only. The current list of supported PWM-enabled driver ICs is as follows:

R² 目前支持 LVCMOS 和 LVDS 模式下的多种类型的 PWM 驱动 IC，以及仅在 LVCMOS 模式下的大多数静态型 IC。当前支持的支持 PWM 的驱动器 IC 列表如下：

- Macroblock MBI5041 (version B)
Macroblock MBI5041 (版本 B)
- Macroblock MBI5042
Macroblock MBI5042 (version B)
Macroblock MBI5042 (B 型)
- Macroblock MBI5043
- Macroblock MBI5050
- Macroblock MBI5051
- Macroblock MBI5051 (version B)
Macroblock MBI5051 (B 版)
- Macroblock MBI5152
- Macroblock MBI5153
- Macroblock MBI5252
- Macroblock MBI6023
- Optotech DR3058 光泰 DR3058
- MY-Semi MY9221 MY-半导体 MY9221
- MY-Semi MY9266 MY-半导体 MY9266
- MY-Semi MY9269 MY-半导体 MY9269
- MY-Semi MY9366

Other types of driver IC can usually be supported on request. Please contact us (supplying a full English-language datasheet) for further information.

通常可以根据要求支持其他类型的驱动器 IC。请联系我们（提供完整的英文数据表）以获取更多信息。

Virtual Output Channels (LVCMOS Mode)

虚拟输出通道（LVCMOS 模式）

Virtual output channels allow each data output on the R² card to drive two chains of driver ICs. This requires a small amount of additional circuitry on the panel's spine/hub board to split and route the signals accordingly. Virtual channels mode is primarily intended for retrofit applications where the existing panel design requires more than the 36 driver IC chains natively supported by R². Wherever possible the panel design should be modified to require 36 (or fewer) chains.

虚拟输出通道允许 R² 卡上的每个数据输出驱动两条驱动器 IC 链。这需要在面板的 spine/hub 板上添加少量额外的电路，以相应地分离和路由信号。虚拟通道模式主要用于现有面板设计需要的 R² 本身支持的 36 个驱动器 IC 链以上的改造应用。应尽可能修改面板设计以需要 36 个（或更少）链。

For example LED_DATA[0] can be routed to two chains providing the two chains are clocked using different DCLKs. An additional signal - LED_DCLK_B is provided for this purpose.

例如，LED_DATA[0] 可以路由到两条链，前提是两条链使用不同的 DCLKs 进行计时。为此，提供了一个附加信号 - LED_DCLK_B。

When splitting signals to feed to two independent chains of driver ICs, it is strongly recommended that the signals are buffered or registered immediately following the split in order to maintain signal integrity. Two additional signals - VIRTUAL_FF_CLK_A and VIRTUAL_FF_CLK_B - are provided to clock flip-flops for registering data into each of the virtual channels. All signal paths to each chain should have the same number of buffering stages to avoid adding skew between the signals.

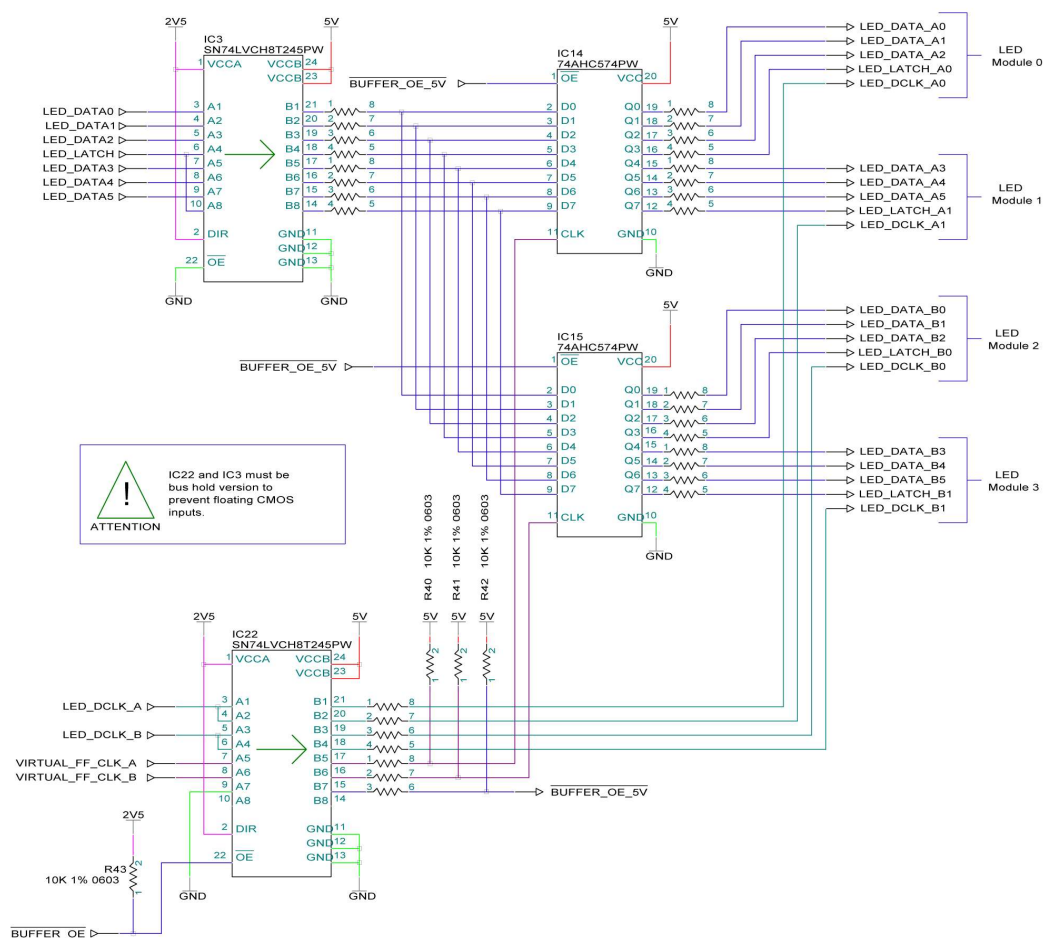
当对信号进行分路以馈送到两个独立的驱动 IC 链时，强烈建议在分路后立即对信号进行缓冲或注册，以保持信号完整性。为时钟 flip-flops 提供了两个额外的信号 - VIRTUAL_FF_CLK_A 和 VIRTUAL_FF_CLK_B - 用于将数据注册到每个虚拟通道中。每个 chain 的所有 signal paths 都应该有相同数量的 buffering stages 以避免在 signals 之间增加 skew。

In panels using PWM-enabled driver ICs the LED_LATCH signal should be split using the same method as the LED_DATA[x] signals. When using static driver ICs the LED_LATCH line should be fanned-out using a buffer. All other signals such as LED_GCLK and LED_SCAN[x] should always be fanned-out using buffers.

在使用使能 PWM 的驱动器 IC 的面板中，LED_LATCH 信号应使用与 LED_DATA[x] 信号相同的方法进行分路。当使用静态驱动器 IC 时，应使用缓冲器将 LED_LATCH 线扇出。所有其他信号，如 LED_GCLK 和 LED_SCAN[x] 应始终使用 buffers 扇出。

A reference design for using virtual channels is shown in [Figure 12](#). For further information and design assistance when using virtual channels please contact us.

使用虚拟通道的参考设计如



Note: This reference design is for PWM-enabled driver ICs. For static-type driver ICs the LED_LATCH signal should not be split using flip-flops.

注：此参考设计适用于支持 PWM 的驱动器 IC。对于静态型驱动器 IC，不应使用 flip-flops 来分割 LED_LATCH 信号。

Note: This reference design only shows using 6 LED_DATA pins to create 12 virtual channels which are split across four LED modules. This should be extrapolated to the required number of virtual channels and modules.

注意：此参考设计仅显示使用 6 个 LED_DATA 引脚创建 12 个虚拟通道，这些虚拟通道分为 4 个 LED 模块。这应该外推到所需的 virtual channels 和 modules 数量。

Figure 12 - Reference design for virtual channels

图 12 - 虚拟通道的参考设计

USER INTERFACE SUPPORT 用户界面支持

The R² supports a range of different user interfaces, such as status LEDs and push buttons. The UI_TEST_BUTTON_N should be held low for >3s to enter/exit test pattern mode and pulsed low briefly to change test patterns. The UI_STATUS[x] pins can be used for connecting status LEDs and other types of user interface.

R² 支持一系列不同的用户界面，例如状态 LED 和按钮。UI_TEST_BUTTON_N 应保持低电平以使 >3s 进入/退出测试模式，并短暂地将脉冲保持低电平以更改测试模式。UI_STATUS[x] 引脚可用于连接状态 LED 和其他类型的用户界面。

Our recommended user interface is a single full-colour RGB LED for indicating status and a single push button for enabling and changing test patterns. These should be connected to the UI_STATUS[x] and UI_TEST_BUTTON_N pins as suggested in [Figure 13](#). The various states indicated on the full-colour RGB LED when using the recommended scheme are documented in [Table 14](#). Note that if the recommend scheme is used then UI_STATUS[0] should be left disconnected.

我们推荐的用户界面是一个用于指示状态的全彩 RGB LED，以及一个用于启用和更改测试模式的按钮。这些应按照 UI_TEST_BUTTON_N UI_STATUS 表 14 - 全彩 RGB 状态 LED 状态指示" data-immersive-translate-walked="59903019-7bf0-413a-ad94-3bf8a89f1cc5">表 14。请注意，如果使用 recommended 方案，则应断开 UI_STATUS[0] 的连接。

Other user interface types and features are supported on a case-by-case basis. Requests should be made by contacting us during the design phase for further details and confirmation of support.

其他用户界面类型和功能将根据具体情况提供支持。应在设计阶段与我们联系以提出请求，以获取更多详细信息和确认支持。

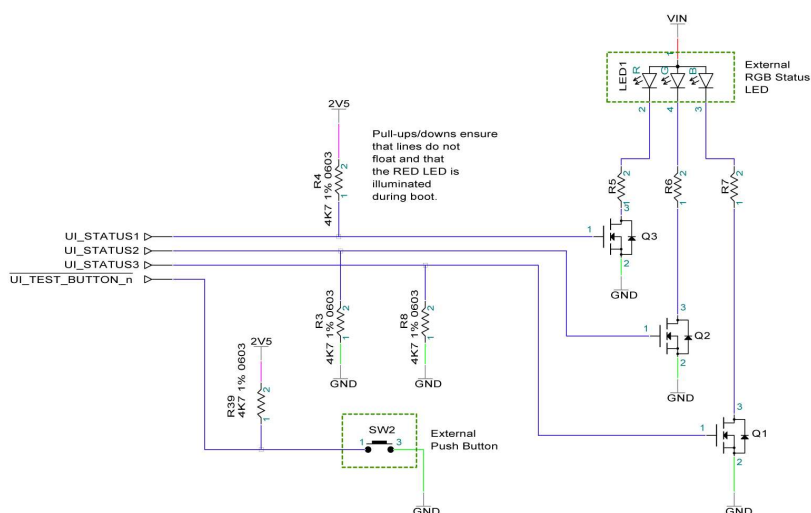


Figure 13 - Recommended user interface circuitry

图 13- 推荐的用户界面电路

Card state 卡片状态	Full-colour RGB LED colour 全彩 RGB LED 颜色
Card powered-down or REBOOT_N asserted 卡已关闭或 REBOOT_N 断言	UI_STATUS[x] lines will be floating, so LED colour will depend on pull-up/-down resistors. We recommend the resistors are arranged as per the reference schematic. UI_STATUS[x] 线将浮动，因此 LED 颜色将取决于上拉/下拉电阻器。我们建议按照参考原理图排列电阻器。
Card rebooting 卡重启	Red solid 红色固体
Error or over-temperature 误差或过热	Red flashing 红灯闪烁
Card booted successfully (no Ethernet links established) 卡已成功启动（未建立以太网链路）	Green solid 绿色固体
One Ethernet link established (not associated to processor) 已建立 1 个以太网链路（未与处理器关联）	Single blue flash 单蓝闪
Two Ethernet links established (not associated to processor) 已建立两条以太网链路（未与处理器关联）	Double blue flash 双蓝闪
Card successfully associated with processor 卡已成功与处理器关联	Blue solid 蓝色固体
Test pattern mode enabled 启用测试模式模式	Yellow solid 黄色固体
Identify/beacon (used to help users identify individual panels) 识别/信标（用于帮助用户识别单个面板）	Yellow flashing 黄色闪烁

Table 14 - Full-colour RGB status LED states

表 14 - 全彩 RGB 状态 LED 状态

SENSORS, FANS AND MONITORING

传感器、风扇和监控

R² supports the monitoring of external sensors, such as temperature sensors and accelerometers. Information from these sensors can then be reported back to users via the user interface of a Tessera Processor. Multiple sensors can be supported, including multiple of the same type. For example, manufacturers may wish to place temperature sensors in various locations across the panel.

R² 支持监控外部传感器，例如温度传感器和加速度计。然后，可以通过 Tessera 处理器的用户界面将来自这些传感器的信息报告给用户。可以支持多个传感器，包括多个相同类型的传感器。例如，制造商可能希望将温度传感器放置在面板的不同位置。

We strongly recommend that at least one temperature sensor is connected to the R² and that this is placed in the same part of the panel's enclosure as the card.

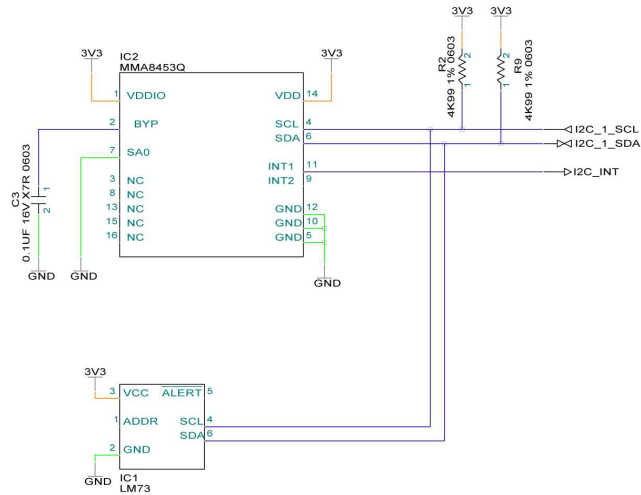
我们强烈建议至少将一个温度传感器连接到 R²，并将其与卡放置在面板外壳的同一部分。

The list of currently supported sensors is given in [Table 15](#). Other types of sensor can usually be supported on request, please contact us for confirmation at the earliest opportunity. A reference design for connecting a temperature sensor and accelerometer is given in [Figure 16](#).

当前支持的传感器列表在 [图 16 - 连接温度传感器和加速度计的参考设计](#) 温度传感器及加速度计连接参考设计 "data-immersive-translate-walked="59903019-7bf0-413a-ad94-3bf8a89f1cc5">图 16.

Part 部分	Type 类型	R ² pins R ² 引脚
Texas Instruments LM73 德州仪器 LM73	Temperature sensor 温度传感器	I2C_1_SDA/_SCL
Sensirion SHT20 森思瑞恩 SHT20	Temperature and humidity sensor 温度和湿度传感器	I2C_1_SDA/_SCL
Freescale MMA8453Q 飞思卡尔 MMA8453Q	Accelerometer (for rotation reporting) 加速度计（用于旋转报告）	I2C_1_SDA/_SCL and I2C_INT I2C_1_SDA/_SCL 和 I2C_INT
Maxim MAX6650	Fan speed regulator and monitor 风扇速度调节器和监视器	I2C_1_SDA/_SCL and I2C_INT I2C_1_SDA/_SCL 和 I2C_INT

Table 15 - Supported sensor parts
表 15 - 支持的传感器部件



Note: the I2C_x_SDA/SCL pins are 3.3V tolerant, so it is not necessary to level-shift the I2C bus.

注意：I2C_x_SDA/SCL 引脚可承受 3.3V 电压，因此无需对 I2C 总线进行电平转换。

Note: pull-up resistors R2 and R9 have a lower resistance than is specified by the I2C standard. This is to ensure the I_{OUT} limit on the I2C pins is not violated. Do not use different values without prior consultation.

注意：上拉电阻 R2 和 R9 的电阻低于 I2C 标准规定的电阻。这是为了确保不违反 I2C 引脚上的 I_{OUT} 限制。未经事先咨询，请勿使用不同的值。

Figure 16 - Reference design for connecting temperature sensor and accelerometer

图 16 - 连接温度传感器和加速度计的参考设计

R² has built-in fan speed control and speed monitoring capabilities. For more information please contact us.

R² 具有内置的风扇速度控制和速度监控功能。欲了解更多信息，请联系我们。

PER-PIXEL CALIBRATION 每像素校准

R² supports the Tessera Calibration System for applying full colour per-LED corrections to make the panel display specific colours and appear more uniform. Calibration data can be stored in the card's internal flash memory, or it can be retrieved from external storage, such as an EEPROM or flash device on each module.

R² 支持 Tessera 校准系统，可对每个 LED 进行全彩校正，使面板显示特定颜色并看起来更均匀。校准数据可以存储在卡的内部闪存中，也可以从外部存储中检索，例如每个模块上的 EEPROM 或闪存设备。

Wherever possible, using storage on the module is preferable as modules can be moved between panels without having to worry about re-programming calibration data. R² supports both EEPROM and flash devices for storage. Both SPI and I2C interfaces are also supported for retrieving the data, though I2C is only recommended for legacy applications.

在可能的情况下，最好在模块上使用存储空间，因为模块可以在面板之间移动，而不必担心重新编程校准数据。R² 支持 EEPROM 和闪存设备进行存储。还支持 SPI 和 I2C 接口来检索数据，但仅建议将 I2C 用于传统应用程序。

These options are described in more detail below.

下面将更详细地介绍这些选项。

External Storage 外部存储

Typically the external storage mode will be used when there is a storage device on each module. However, it could also be used in cases where it is not possible to put devices on the modules themselves, but the pixel count is too high to use internal storage mode.

通常，当每个模块上都有存储设备时，将使用外部存储模式。但是，它也可以在无法将设备放在模块本身上，但像素数太高而无法使用内部存储模式的情况下使用。

Both EEPROM and flash devices are supported for external storage. EEPROM devices are usually preferable due to their simplicity, but large capacity parts are generally not available, in which case a flash device should be used. The list of currently supported device families is given in [Table 17](#). Other parts can sometimes be supported on request - please contact us for confirmation at the earliest opportunity.

EEPROM 和闪存器件都支持外部存储。EEPROM 器件通常因其简单性而更可取，但通常没有大容量器件，在这种情况下，应使用闪存器件。当前支持的设备系列列表在

Device type 设备类型	Supported parts 支持的部件
SPI Flash SPI 闪存	Micron M25P or Winbond W25X series 美光 M25P 或华邦 W25X 系列
SPI EEPROM	Atmel AT25 series (or equivalent) Atmel AT25 系列（或同等产品）
I2C EEPROM	Atmel AT24C series (or equivalent) Atmel AT24C 系列（或同等产品）

Table 17 - Supported storage devices
表 17 - 支持的存储设备

The recommended max pixel count for a given size of storage device is given in [Table 18](#). This refers to the recommended maximum number of pixels per device, so if there is a storage device per module then this should be considered the pixel count limit for the module, not the entire panel. These recommendations are based on storing up to two calibrations, each containing both correction coefficients and raw measurements, assuming EEPROM is used for device sizes 2Mb and below. If you have specific requirements for calibration please contact us to discuss them as in some cases it may be advisable to use a larger storage device than those listed.

给定大小的存储设备的推荐最大像素数在

Storage device size 存储设备大小	Recommended max pixel count per device 建议每个设备的最大像素数
16Mb 16兆字节	25000
8Mb 8兆字节	12000
4Mb 4兆字节	6000
2Mb 2兆字节	3000
1Mb 1兆字节	1500
512Kb 512兆字节	700
256Kb 256兆字节	300
128Kb 128兆字节	150

Table 18 - Recommended storage device sizes

表 18 - 建议的存储设备大小

In LVCMOS mode, R² interfaces with external devices using the CAL_I2C_SPI[x] pins. These are mapped to an I2C or SPI interface as described in [Table 19](#). For new designs, especially with high pixel counts, SPI interfaces are recommended as they are higher bandwidth than I2C, which reduces boot times. SPI also tends to be more reliable than I2C. [Figure 20](#) shows an example circuit for connecting these pins to four SPI storage devices, for example for panels with four modules. If you are intending to use other modes, including I2C, please contact us for more information and a reference design.

在 LVCMOS 模式下，R² 使用 CAL_I2C_SPI[x] 引脚与外部器件连接。这些映射到 I2C 或 SPI 接口，如 [图 20 - 使用 SPI 连接四个校准器件的参考设计 4 组校正数据存储设备连接设计参考](#) "data-immersive-translate-walked="59903019-7bf0-413a-ad94-3bf8a89f1cc5">图 20 显示了将这些引脚连接到四个 SPI 存储设备的示例电路，例如具有四个模块的面板。如果您打算使用其他模式，包括 I2C，请联系我们以获取更多信息和参考设计。

For panels with more than four modules, up to sixteen SPI buses can be supported using a small external multiplexing circuit. Please contact us to request a reference design.

对于具有 4 个以上模块的面板，使用小型外部多路复用电路可支持多达 16 个 SPI 总线。请联系我们索取参考设计。

In LVDS mode, the LVDS links to each module are used to communicate with external storage devices. LVDS Mode is not yet available on general release. If you are interested in using LVDS mode in your project please contact us for further information.

在 LVDS 模式下，指向每个模块的 LVDS 链接用于与外部存储设备通信。LVDS 模式在通用版本中尚不可用。如果您有兴趣在项目中 使用 LVDS 模式，请联系我们以获取更多信息。

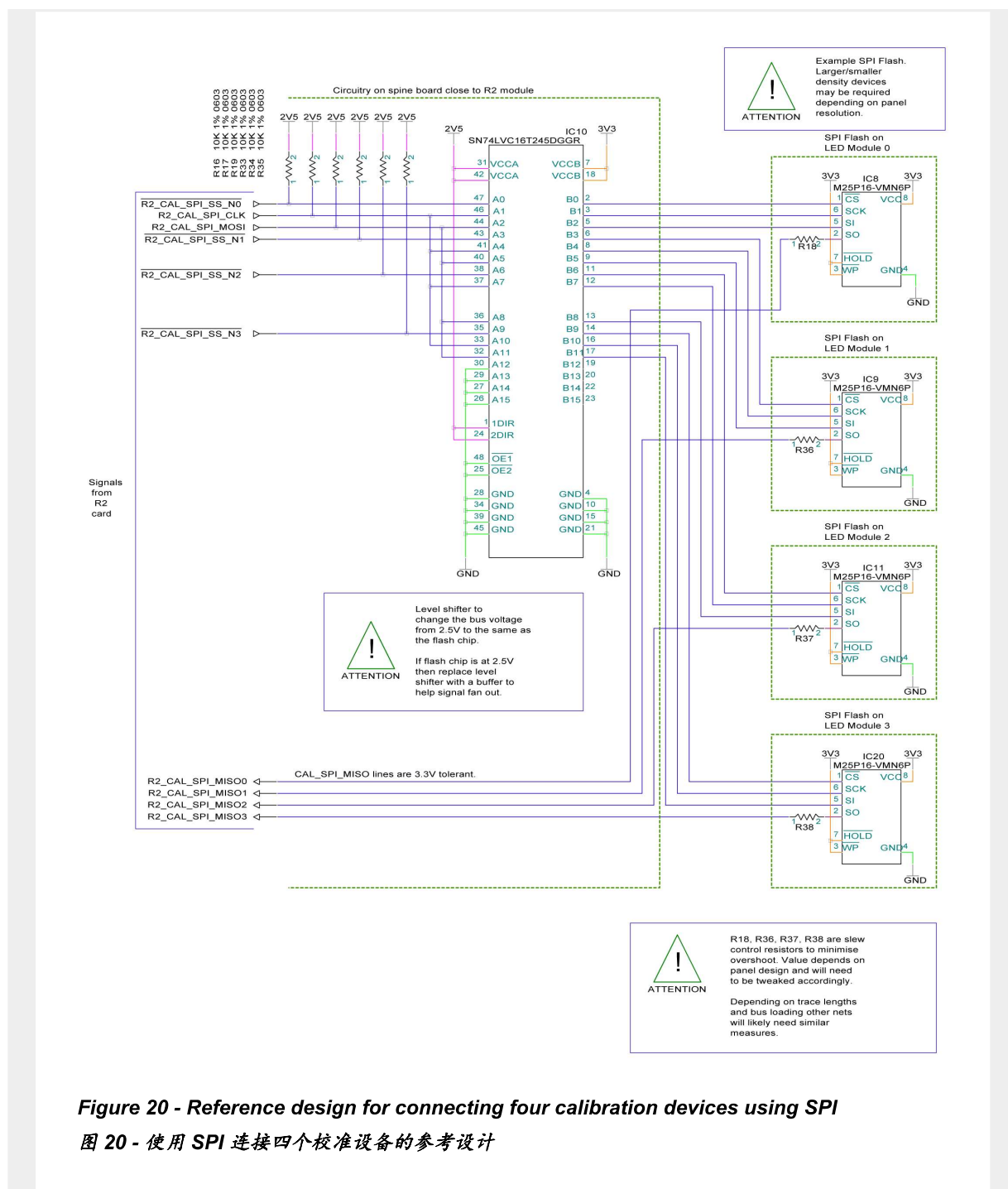
R ² pin R ² 引脚	SPI four module interface SPI 四模块接口	SPI external multiplexer interface SPI 外部多路复用器接口	I2C four module interface I2C 四模块接口	I2C external multiplexer interface I2C 外部多路复用器接口	UART interface U ART 接口
CAL_I2C_SPI[0]	MOSI	MOSI	RESERVED - do not connect RESERVED - 不连接	RESERVED - do not connect RESERVED - 不连接	RESERVED - do not connect RESERVED - 不连接
CAL_I2C_SPI[1]	CLK	CLK	SDA_OUT[3]	DIR	RESERVED - do not connect RESERVED - 不连接
CAL_I2C_SPI[2]	MISO[0] 味噌[0]	MISO	SCL_OUT	SCL_OUT	RX[0] 接收 [0]
CAL_I2C_SPI[3]	SS_N[0]	SS_N	SDA_IN[0]	SCL_IN	TX[0]
CAL_I2C_SPI[4]	MISO[1] 味噌[1]	RESERVED - do not connect RESERVED - 不连接	SDA_OUT[0]	SDA_OUT	RX[1] 接收 [1]
CAL_I2C_SPI[5]	SS_N[1]	MUX[4] 多路复用器[4]	SDA_IN[1]	SDA_IN	TX[1]
CAL_I2C_SPI[6]	MISO[2] 味噌[2]	MUX[0] 多路复用器[0]	SDA_OUT[1]	SDA_MUX[0]	RX[2] 接收 [2]
CAL_I2C_SPI[7]	SS_N[2]	MUX[1] 多路复用器[1]	SDA_IN[2]	SDA_MUX[1]	TX[2] 德克萨斯州[2]
CAL_I2C_SPI[8]	MISO[3] 味噌[3]	MUX[2] 多路复用器[2]	SDA_OUT[2]	SDA_MUX[2]	RX[3] 接收 [3]
CAL_I2C_SPI[9]	SS_N[3]	MUX[3] 多路复用器[3]	SDA_IN[3]	SDA_MUX[3]	TX[3] 德克萨斯州[3]

Note: I2C and UART modes should not be used without prior consultation.

注意：未经事先咨询，不得使用 I2C 和 UART 模式。

Table 19 - Mapping of CAL_I2C_SPI[x] pins to SPI, I2C or UART interfaces (LVCMOS mode only).

表 19 - CAL_I2C_SPI[x] 引脚到 SPI、I2C 或 UART 接口的映射（仅限 LVCMOS 模式）。



Internal Storage 内部存储

If no external storage is available, it is possible to store calibration data in the R² card's internal flash. The storage space available is limited, and therefore it may not be possible to store multiple calibration banks in internal storage. Internal storage does not currently support storing calibration data for more than 16384 pixels.

如果没有可用的外部存储，则可以将校准数据存储在 R² 卡的内部闪存中。可用的存储空间有限，因此可能无法在内部存储器中存储多个校准库。内部存储目前不支持存储超过 16384 像素的校准数据。

DESIGN PROCESS 设计过程

We are happy to partner with panel manufacturers directly or via an end customer such as rental company.

我们很乐意直接与面板制造商合作，或通过租赁公司等最终客户合作。

During the design process there should be an ongoing dialogue between us and our partners, usually including direct communication with the panel manufacturer. This is to ensure that our partners get the results they want from the Tessera processing and it enables us to ensure that the Tessera brand remains a mark of quality for LED video processing.

在设计过程中，我们与合作伙伴之间应该进行持续的对话，通常包括与面板制造商的直接沟通。这是为了确保我们的合作伙伴从 Tessera 处理中获得他们想要的结果，并使我们能够确保 Tessera 品牌仍然是 LED 视频处理的质量标志。

We respect our partners' need for confidentiality and we ask our partners not to disclose proprietary information about our technology. Sometimes it will be necessary to enter into a mutual non-disclosure agreement (NDA) with partners. If a partner would like to be covered by an NDA then they should notify us at the earliest opportunity.

我们尊重合作伙伴的保密需求，并要求合作伙伴不要披露有关我们技术的专有信息。有时有必要与合作伙伴签订相互保密协议（NDA）。如果合作伙伴希望获得 NDA，则应尽早通知我们。

A typical design process would be as follows:

典型的设计过程如下：

Initiation 开始

It is essential that partners make contact with us at the start of the design process. Partners should tell us as much as possible about the product into which Tessera processing is being integrated so that we can best support the design process. We can then make any specific recommendations which may differ from the general guidance offered in this document.

合作伙伴在设计过程开始时与我们取得联系至关重要。合作伙伴应尽可能多地告诉我们集成了 Tessera 处理的产品，以便我们能够最好地支持设计过程。然后，我们可以提出任何可能与本文档中提供的一般指南不同的具体建议。

Design Creation and Review

设计创作和审查

Partners should use the technical information in this document to integrate Tessera processing into their design. We encourage partners to contact us should any clarification or additional guidance be required. In some sections of this document it is requested that partners contact us to confirm that there is support for certain features, for example particular external peripherals. It is important that this is done at the earliest opportunity.

合作伙伴应使用本文档中的技术信息将 Tessera 处理集成到他们的设计中。如果需要任何说明或其他指导，我们鼓励合作伙伴与我们联系。在本文档的某些部分，要求合作伙伴与我们联系，以确认支持某些功能，例如特定的外部外围设备。尽早完成此作非常重要。

Once the design is complete we are happy to review schematics for incompatibilities. This service is usually complementary and subject to the availability of engineers' time.

设计完成后，我们很乐意审查原理图的不兼容性。这项服务通常是互补的，并取决于工程师的时间。

Prototyping 成型

After schematic review partners should proceed to produce a small number of prototype panels. During this time we will create a Configuration for the panel. This is the file that tells the R² card how to control the panel, and includes information such as the type of driver IC and the ordering of the LEDs. We ask that a couple of these prototypes are then sent to us so that we can test the Configuration and tune accordingly.

原理图审查后，合作伙伴应继续生产少量原型面板。在此期间，我们将为面板创建一个 Configuration。该文件告诉 R² 卡如何控制面板，包括驱动器 IC 的类型和 LED 的顺序等信息。我们要求将其中的几个原型发送给我们，以便我们可以测试 Configuration 并相应地进行调整。

If necessary we are often able to provide 'light-up' firmware so that partners can run basic tests on the panel before sending them to us.

如有必要，我们通常能够提供“light-up”固件，以便合作伙伴可以在将面板发送给我们之前在面板上运行基本测试。

Once Configuration testing is complete we will supply these files and tools to the partner so that they can program the R² card and complete their prototype validation and testing.

配置测试完成后，我们将向合作伙伴提供这些文件和工具，以便他们可以对 R² 卡进行编程并完成原型验证和测试。

Production and Ongoing Support

生产和持续支持

We ask that partners let us retain one or two panels (or supply us with production panels if sufficiently different) so that we can provide ongoing support to our end-users. The Tessera software and firmware is constantly evolving as new features are added, so this also lets us check that new features work correctly on all partners' panels.

我们请求合作伙伴允许我们保留一两个面板（如果差异足够大，则向我们提供生产面板），以便我们可以为最终用户提供持续的支持。随着新功能的添加，Tessera 软件和固件也在不断发展，因此这也让我们可以检查新功能是否在所有合作伙伴的面板上都能正常工作。

If calibration is required we will discuss the options with partners and assist them in establishing appropriate manufacturing and calibration procedures.

如果需要校准，我们将与合作伙伴讨论选项，并协助他们建立适当的制造和校准程序。

SALES AND SUPPORT 销售和支持

For further information or to purchase Tessera processing, please contact us.

如需更多信息或购买 Tessera 处理服务，请联系我们。

Email: info@bromptontech.com

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REVISION HISTORY 修订历史记录

Revision 校订	Date 日期	Comments 评论
1	22 September 2014 22 九月 2014	Preliminary release for internal review 供内部审查的初步发布
2	9 October 2014 9 十 月 2014	First release to customers 向客户发布的第一个版本
3	17 October 2014 17 十月 2014	Adding virtual channels reference design 添加虚拟通道参考设计
4	11 December 2014 11 十二月 2014	Updating supported driver ICs. 更新支持的驱动程序 IC。 Recommending that reverse-mounted connectors are not used. 建议不要使用反向安装的连接器的。
5	19 December 2014 19 十二月 2014	Added scan multiplex ratios up to 32:1 增加了高达 32: 1 的扫描多路复用率
6	27 January 2015 27 一月 2015	Added pin options for LED_DATA[35:32] 添加了 LED_DATA 的引脚选项 [35: 32]
7	27 February 2015 27 二月 2015	Updated Figure 11 , Figure 12 and Figure 20 . 更新了 图 12 - 虚拟通道参考设计 虚拟通道参考设计 “data-immersive-translate-walked=”59903019-7bf0-413a-ad94-3bf8a89f1cc5”>图 12 和 Added UART mode to Table 19 . 在
8	9 July 2015 9 七月 2015	Formatting changes only - no changes to content. 仅格式更改 - 不对内容进行更改。
9	10 July 2015 10 七 月 2015	Added subscription information. 添加了订阅信息。
10	17 August 2015 17 八月 2015	Changed location of LED_DATA[34] and LED_DATA[35] pins to enable simultaneous use with LED_SCAN[3] and LED_SCAN[4]. 更改了 LED_DATA[34] 和 LED_DATA[35] 引脚的位置, 以便与 LED_SCAN[3] 和 LED_SCAN[4] 同时使用。 Removed FAN_CONTROL and FAN_STATUS pins, and added RESERVED_GPIO[x] pins. 删除了 FAN_CONTROL 和 FAN_STATUS 引脚, 并添加了 RESERVED_GPIO[x] 引脚。 Minor text corrections and clarifications. 细微的文本更正和澄清。
11	14 September 2015 14 九月 2015	Correcting typographical error. 更正印刷错误。
12	12 April 2016 12 四 月 2016	Adding Chinese translation, updating figures and fixing typographical errors. 添加中文翻译、更新图表和修复印刷错误。 Updating supported driver IC list.

更新支持的驱动程序 IC 列表。

Adding debug header. 添加调试标头。

Adding new I2C four module mode.

添加新的 I2C 四模块模式。

Shortened calibration function names for readability.

缩短了校准函数名称以提高可读性。

Calibration text clarifications.

校准文本说明。

Updated [Table 18](#) to allow for new OSCA feature.

更新了